Included Variable Bias in Dynamic Panels: A Diagnostic for Time-Varying Controls and a Benchmark for the Contemporaneous Treatment Effect

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Abstract

We study included variable bias as a general diagnostic problem that arises when researchers condition on observed covariates that may themselves respond to treatment. Recent work on difference-in-differences with time-varying covariates shows that in such settings both conditioning on the realized covariate and omitting it can be problematic, because the causally relevant object may resemble an untreated potential covariate that is not directly observed. We derive a closed-form included variable bias diagnostic for linear nested models, $\text{IVB} = -\theta^{\star}\pi$, and show how directed acyclic graphs determine whether the resulting specification shift reflects collider bias, over-control, or deconfounding. We then develop the paper's main applied implication: in time-series cross-sectional settings focused on the contemporaneous treatment effect, ADL + FE with lagged state variables provides a practical benchmark. Simulations covering collider, dual-role, and mediator designs show that this benchmark reduces bias in the reported designs and preserves the contemporaneous total effect when lagged rather than contemporaneous controls are used. Applications illustrate how IVB quantifies specification shifts while the benchmark provides a defensible starting point for applied work.

1 Introduction

Researchers using time-series cross-sectional data constantly face a deceptively simple specification choice: should an observed time-varying covariate be included in the regression? Consider an international-relations

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scholar studying the effect of UN peacekeeping on democratization. Conditioning on conflict intensity is tempting because more severe conflicts are more likely to receive peacekeeping missions and are less likely to democratize. But peacekeeping can itself reduce violence. If so, conditioning on contemporaneous conflict intensity means conditioning on a variable that may already have responded to treatment. Omitting conflict may leave confounding; including realized conflict may introduce a different problem. The same logic appears whenever researchers want to control for GDP, aid, inflation, turnout, repression, or any other variable that evolves within the same period as treatment and outcome.

Recent work on difference-in-differences with time-varying covariates makes this problem stark. When a covariate can itself respond to treatment, both conditioning on the realized covariate and omitting it can be problematic, because the causally relevant object may be something like the untreated potential covariate $Z_t(0)$ rather than the realized Z_t (Caetano and Callaway, 2024; Caetano et al., 2022; Lin and Zhang, 2022). Related work on panels and causal dynamics likewise shows that time-varying covariates can create demanding identification problems (Blackwell and Glynn, 2018; Imai and Kim, 2021). We take that broader literature as the starting point rather than the target of critique.

Our contribution is different and narrower, but also more operational. We introduce included variable bias (IVB) as a general diagnostic problem that arises when researchers condition on observed covariates that may themselves respond to treatment. In linear nested models, the resulting specification shift admits a simple closed-form decomposition:

$$\widehat{\beta}^{\star} - \widehat{\beta} = -\widehat{\theta}^{\star}\widehat{\pi},$$

where $\widehat{\theta}^{\star}$ is the coefficient on the candidate control in the long regression and $\widehat{\pi}$ is the coefficient on treatment in an auxiliary regression for that control. Directed acyclic graphs (DAGs) tell the researcher what kind of variable Z is; IVB tells the researcher how much including it moves the treatment coefficient. This is the paper's general contribution. The current manuscript develops it for the linear models most common in TSCS work, where the diagnostic is exact and directly estimable.

The paper's main applied payoff is narrower. For TSCS settings focused on the contemporaneous treatment effect (CET), the IVB logic and the associated timing argument imply a practical benchmark: start from ADL + FE with lagged state variables, use lagged covariates when substantively justified, and treat contemporaneous time-varying controls as suspect until the timing is explicit. This is not the claim that ADL solves every panel-data problem. It is the claim that, for the CET, ADL + FE provides a defensible baseline when controlling correctly for time-varying covariates would otherwise require stronger and often untestable assumptions.

This architecture helps distinguish the paper from two nearby contributions. It is not only a warning that time-varying controls are hard, because IVB gives that difficulty a named and estimable diagnostic. And it is not only a recommendation to use dynamic specifications, because the benchmark is presented as the main applied implication of a more general included-variable-bias framework. The broader research program behind IVB may well travel beyond the current setting, but this paper develops its full payoff in the linear TSCS case.

We proceed in four steps. Section~2 defines the control-variable problem when observed covariates may themselves respond to treatment. Section~3 uses DAG logic to show why the same substantive variable can be harmful at t and useful at $t-1$. Section~4 presents IVB as the paper's general diagnostic contribution and derives the closed-form decomposition used in linear TSCS models. Section~5 then develops the main applied implication of that framework: a benchmark for the CET based on ADL + FE with lagged state variables. Section~6 shows how IVB and the benchmark interact in published TSCS studies. Section~7 closes with the paper's claim, its limits, and the agenda it opens.

2 The Time-Varying Control Problem for the Contemporaneous Treatment Effect

Applied researchers working with time-series cross-sectional (TSCS) data rarely face a binary choice between control'' anddo not control.' ' The more realistic problem is whether an observed time-varying covariate should enter the specification when the estimand is the contemporaneous treatment effect (CET), that is, the effect of D_t on Y_t. The same variable can help, hurt, or redefine the estimand depending on timing. A contemporaneous covariate Z_t may be a confounder, a collider, or a mediator. Its lag, Z_{t-1}, may instead behave as a predetermined state variable relative to the CET.

Standard heuristics are too coarse for this problem. Control checking'' recommends adding predictors of the outcome to gain precision;confounding checking' ' recommends adding joint causes of treatment and outcome (Cinelli, Forney and Pearl, 2022). Both heuristics ignore the possibility that a candidate control is post-treatment in the period of interest. That omission is costly in TSCS designs because treatment, outcome, and covariates evolve jointly over time, often with persistence in all three processes (Blackwell and Glynn, 2018; Imai and Kim, 2021).

Recent DID work sharpens the problem further: if Z_t may itself respond to treatment, neither conditioning on realized Z_t nor omitting it necessarily identifies the target estimand, because the relevant object may be the untreated potential covariate $Z_t(0)$ rather than the observed Z_t (Caetano and Callaway, 2024; Caetano et al., 2022). We do not solve that identification problem in general. Instead, we ask a more operational question. When applied researchers condition on an observed time-varying covariate that may respond to treatment, how much does that choice move the treatment coefficient, and what is the safest default specification in the TSCS case?

Our answer has two parts. The general contribution is IVB: a named diagnostic problem and an exact decomposition for linear nested models. The more specific contribution is the paper's payoff in TSCS settings with the CET: a dynamic benchmark based on ADL + FE with lagged state variables. In other words, the paper is not only about a safer specification rule and not only about a formula. It is about how the formula makes the safer rule operational.

The scope qualification matters. A lagged control can be called pre-treatment only relative to the effect of D_t on Y_t, and only when its substantive timestamp supports that ordering. Even then, it need not be innocuous for cumulative, delayed, or long-run effects. The paper's positive recommendation is therefore explicitly local to the CET. Difference-in-differences and related panel designs remain part of the motivation for why time-varying controls are hard, but the paper's formal and applied results are developed in the linear TSCS setting.

3 DAGs, Timing, and Why Z_t Is Not Z_{t-1}

The causal role of a time-varying covariate depends on its substantive timing, not its subscript alone. Before classifying a control, we therefore need to map the variables recorded in a panel row onto a causal sequence within the period.

3.1 The CET and the Within-Period Causal Clock

Let H_{it}^- denote the information and state variables fixed immediately before the period-t treatment begins. For two well-defined treatment levels d and d', we define the contemporaneous treatment effect as

$$\tau_{\text{CET}}(d, d'; h) = \mathbb{E}\left[Y_{it}(d) - Y_{it}(d') \mid H_{it}^- = h\right],$$

where $Y_{it}(d)$ is measured after the period-t intervention and any within-period responses it induces. Thus, unless a controlled direct effect is explicitly targeted, the CET includes paths such as $D_{it} \to Z_{it} \to Y_{it}$. In a constant linear model with a scalar treatment, β_{CET} is the corresponding unit contrast, or marginal effect for a continuous dose.

The substantive clock underlying this definition is

$$H_{it}^{-} \longrightarrow D_{it} \longrightarrow Z_{it} \longrightarrow Y_{it}^{+},$$

where H_{it}^{-} is the state at the opening of the causal period, D_{it} records the onset and intensity of treatment, Z_{it} is allowed to respond, and Y_{it}^{+} is measured after the relevant exposure window. This ordering is an empirical claim, not a consequence of notation. Table~1 states the minimum timestamp information needed to assess it.

This displayed chain is the candidate clock when Z_t is a mediator. It is not imposed on every graph below. A contemporaneous-collider claim $D_t \to Z_t \leftarrow Y_t$ instead requires the outcome process represented by Y_t to precede the measurement of Z_t within the causal period, while D_t can affect both. Thus the common subscript denotes a panel period, not simultaneity or a universal within-period order; each application must defend the mechanism-specific clock.

Table 1: Minimum timestamp record for defining the contemporaneous treatment effect. A lag label is not evidence that a covariate is pretreatment.

Variable	Minimum.timestamp.record	Requirement.for.the.CET.clock
D	Onset, end, and intensity window; whether the measure is a stock, flow, event, or period average	Defines the intervention or exposure contrast within period t
Z	Reference and measurement window; whether it can respond to anticipated, prior, or current treatment	Must close before the first relevant exposure to be called pretreatment; otherwise classify its timing as plausible or ambiguous
Y	Reference window and measurement time; whether any part is recorded before treatment begins	Must follow the relevant exposure window for the stated CET; overlapping aggregates require a finer estimand

Three complications are common in annual and other aggregated panels. First, anticipation can make a recorded Z_{t-1} treatment-responsive before the formal onset of D_t. Second, when treatment varies continuously within the period, a scalar annual value may summarize a treatment path with repeated $D \leftrightarrow Z \leftrightarrow Y$ feedback; the relevant intervention is then a contrast between exposure trajectories, not automatically a one-unit change in the annual average. Third, annual measures may overlap: a late-year treatment can occur after much of Z_t or Y_t has already been realized, while a beginning-of-year outcome may precede the nominally contemporaneous treatment. When documentation establishes nonoverlapping windows, timing is known; when institutional knowledge supports but does not timestamp the sequence, it is plausible; and when windows overlap or remain undocumented, it is ambiguous. The lagged-control benchmark below applies only after this clock has been defended.

3.2 Dynamic Causal Structures for the CET

The elementary fork, chain, and collider remain the building blocks of causal graphs (Pearl, 2009; Elwert and Winship, 2014), but TSCS specifications combine those structures across periods. Figure 1 makes the inherited path explicit for a contemporaneous collider. The blue arrows mark the treatment–outcome causal edges in successive periods. Conditioning on Z_t opens $D_t \to Z_t \leftarrow Y_t$. Because Z_t is also a descendant of the lagged collider Z_{t-1}, that conditioning can activate the inherited path $D_t \leftarrow D_{t-1} \to Z_{t-1} \leftarrow Y_{t-1} \to Y_t$. Conditioning on Y_{t-1} blocks this inherited path at a noncollider, but it does not close the contemporaneous path opened at Z_t. The graph also shows the ordinary dynamic path $D_t \leftarrow D_{t-1} \to Y_{t-1} \to Y_t$; that path is distinct from collider activation and is likewise blocked at Y_{t-1}.

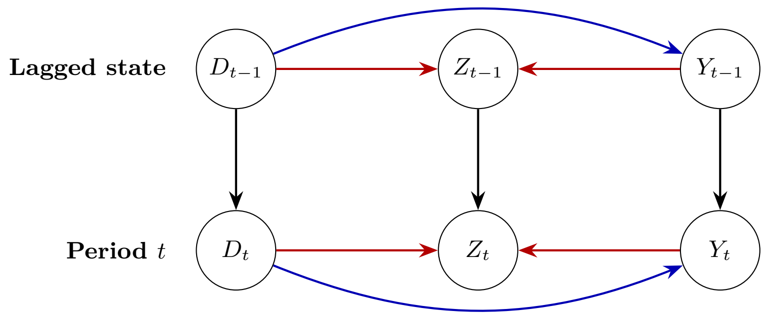

Figure 1: Dynamic contemporaneous-collider DAG for the CET. Red arrows meet at Z_t and Z_{t-1}; blue marks the causal $D_t \to Y_t$ path. Conditioning on Z_t opens the contemporaneous collider and, as conditioning on a descendant of Z_{t-1}, can also activate the inherited lag path.

Figure 2 represents a different role for the same contemporaneous variable. Here Z_t transmits part of the effect of D_t to Y_t, while Z_{t-1} is a lagged state that can affect current treatment and outcome. For the total CET, both current-period blue paths belong to the estimand. Conditioning on a substantively pretreatment Z_{t-1} blocks $D_t \leftarrow Z_{t-1} \to Y_t$ without blocking the within-period mediated path. The remaining lag path $D_t \leftarrow D_{t-1} \to Y_{t-1} \to Y_t$ requires D_{t-1}, Y_{t-1}, or other sufficient history. Conditioning on Z_t instead blocks $D_t \to Z_t \to Y_t$ and changes the target to a direct-effect parameter under additional mediation assumptions.

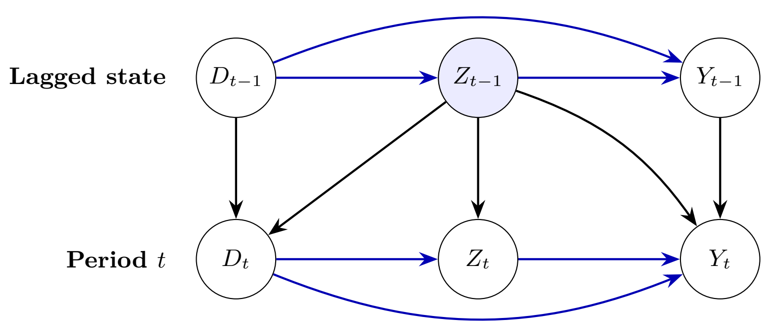

Figure 2: Dynamic contemporaneous-mediator DAG with lagged confounding and state dependence. For the total CET, the direct path and the path through Z_t remain open. Conditioning on a genuinely pretreatment Z_{t-1} can block lagged confounding; conditioning on Z_t blocks the displayed mediated path.

3.3 Timing Logic for Candidate Controls

Table~2 summarizes the main timing logic. The key point is that the paper's benchmark is anchored to the CET rather than to a generic slogan such as "lag everything.' '

Table 2: Timing logic for observed controls when the estimand is the contemporaneous treatment effect. Subscripts identify candidate roles; the substantive clock determines the causal classification.

Candidate.control	Possible.role.for.the.CET	Main.risk.or.benefit	Default.action
Z_t	Collider if post-treatment	Opens a noncausal collider path	Do not condition unless the DAG and causal clock rule out post-treatment status
Z_t	Mediator if post-treatment	Blocks part of the total effect	Do not condition if the target is the total contemporaneous effect
Z_{t-1}	Predetermined state if timestamp passes	Captures state dependence	Candidate benchmark control only after the timestamp check
Z_{t-1}	Lagged confounder if timestamp passes	Removes OVB without blocking the contemporaneous path	Candidate benchmark control only after the timestamp check

Three cases are especially relevant for the remainder of the paper. First, conditioning on a contemporaneous collider Z_t opens a noncausal collider path and can create IVB. Second, a contemporaneous mediator Z_t creates over-control if the target is the total contemporaneous effect. Third, a dual-role variable can be both a lagged confounder and a contemporaneous collider. In that case, the empirical question is not whether the variable should ever appear in the specification, but whether its documented measurement window places it before or after the relevant treatment begins.

4 The Included Variable Bias Diagnostic

The paper's general contribution is to define included variable bias as a distinct diagnostic problem. Once researchers accept that an observed control may itself respond to treatment, they need a language for the resulting specification shift. DAGs classify the control. IVB quantifies what that classification means for the treatment coefficient in a given linear specification.

Consider the short model

$$Y = \beta D + W\delta + u \tag{1}$$

and the long model

$$Y = \beta^\star D + \theta^\star Z + W\delta^\star + u^\star, \tag{2}$$

where W collects admissible controls. By the Frisch–Waugh–Lovell theorem,

$$\widehat{\beta}^\star - \widehat{\beta} = -\widehat{\theta}^\star \widehat{\pi}, \tag{3}$$

where $\widehat{\pi}$ is the coefficient on D in the auxiliary regression of Z on D and W. In cross-sections, W may contain baseline covariates. In TSCS settings, W can include fixed effects, lagged dependent variables, lagged treatment, and other lagged state variables. The formula is therefore an exact linear-model identity, not a separate estimator. That exactness is what makes IVB useful as more than a metaphor: the diagnostic can be read directly off nested linear regressions already familiar to applied researchers.

Two interpretive points matter. First, the formula is diagnostic rather than self-justifying. If Z is a confounder, the same arithmetic difference reflects deconfounding rather than harm. If Z is a mediator, the difference reflects movement from the total to the direct effect (Acharya, Blackwell and Sen, 2016). Only the DAG tells the researcher which interpretation applies. Second, the formula remains useful precisely because it converts a vague modeling dispute into two estimable components: how strongly Z predicts the outcome in the long model, and how strongly D predicts Z once admissible controls have been partialled out.

This is also where the paper's originality lies. Work on dynamic identification, sequential ignorability, and DID with time-varying covariates shows why observed controls may be problematic (Blackwell and Glynn, 2018; Imai and Kim, 2021; Caetano and Callaway, 2024). IVB contributes a different object: a named and estimable decomposition of the specification shift itself. The current paper develops that object in the linear TSCS domain, where it yields both a diagnostic and, in the next section, a practical benchmark. Whether analogous IVB diagnostics can be built for synthetic-control-style estimators, synthetic DID, or factor models is an important agenda, but not a result claimed here.

For the CET in TSCS settings, the practically relevant auxiliary regression is dynamic. If the benchmark specification is

$$Y_{it} = \beta D_{it} + \rho Y_{i,t-1} + \kappa D_{i,t-1} + X'_{i,t-1}\lambda + \alpha_i + \tau_t + \varepsilon_{it}, \tag{4}$$

then the corresponding IVB decomposition asks how much the treatment coefficient shifts when a candidate control is moved from the lagged-state set $X_{i,t-1}$ into the contemporaneous set. This is the operational meaning of "diagnosing bad controls' ' in the remainder of the paper.

The formula also clarifies a point that is easy to miss in practice: lagging the treatment is not a mechanical cure. If D_t is autocorrelated, a collider that is contemporaneous with D_t can remain associated with D_{t-1}. What matters is not whether the treatment index moved backward by one period, but whether the specification blocks the transmission of the bad-control path for the estimand of interest.

4.1 When ADL + FE Identifies the CET

The ADL + FE benchmark identifies the CET only under explicit scope conditions. Let H^-_{it} be the observed pre-exposure history defined above. A calendar-lag vector $H_{i,t-1}$ is a valid implementation only when the timestamp checklist establishes $H_{i,t-1} \subseteq H^-_{it}$ and the included lags and transformations are sufficient for current assignment and the conditional outcome mean. A lag label alone establishes neither condition. A sufficient constant-coefficient restriction is

$$\mathbb{E}[Y_{it}(d) \mid H_{it}^{-}, \alpha_i, \tau_t] = \alpha_i + \tau_t + r(H_{it}^{-})'\gamma + \beta_{\text{CET}} d,$$

together with consistency, no interference, no anticipation, sequential mean exchangeability of D_{it} given H_{it}^{-}, and conditional positivity for the treatment contrast. For discrete treatment, the target levels must have positive conditional probability; for continuous treatment, the target doses must lie in the conditional common support with local support around them. The causal clock must also be recursive: D_{it} precedes the contemporaneous outcome innovation, so simultaneous $Y_{it} \to D_{it}$ feedback is excluded. By contrast, $Y_{i,t-1} \to D_{it}$ feedback is compatible with identification when $Y_{i,t-1} \in H_{it}^{-}$ and the conditioned history is sufficient.

These causal support conditions are distinct from the rank condition for the population projection. If $\widetilde{D}_{it}$ is the population residual from projecting D_{it} on the included history and the unit and time effects, and $0 < \mathbb{E}[\widetilde{D}_{it}^2] < \infty$, then

$$\beta_D^P = \frac{\mathbb{E}[\widetilde{D}_{it} Y_{it}]}{\mathbb{E}[\widetilde{D}_{it}^2]} = \beta_{\text{CET}}.$$

The online appendix states this result as a proposition and gives the full proof. This equality is a population identification result. Unit effects absorb additive time-invariant unit factors and time effects absorb additive common shocks; neither removes omitted time-varying confounding or heterogeneous responses to common shocks. A misspecified lag order or nonlinear history also breaks the stated conditional mean. If treatment effects or dynamics vary over units, periods, or histories, the single coefficient is generally a projection-weighted summary rather than a common CET.

Table 3: Scope conditions for the ADL + FE benchmark: identification versus estimation and inference.

Layer	Scope condition	Interpretation when the condition fails
Population identification	Consistency; no interference; sequential mean exchangeability given the pre-exposure history; no unobserved contemporaneous confounding	The ADL coefficient need not equal the CET
History and timing	No anticipation; $H_{i,t-1} \subseteq H_{it}^{-}$ with sufficient included history; no simultaneous $Y_{it} \to D_{it}$ feedback	A nominal lag may be post-treatment, or omitted history and simultaneity may invalidate the comparison
Conditional support	The target treatment levels lie in the conditional support; continuous doses have local common support	The causal contrast requires unsupported extrapolation
Projection rank	$0 < \mathbb{E}[\widetilde{D}_{it}^2] < \infty$ after partialling out included history and fixed effects	The population FWL coefficient is not well defined even if causal overlap holds for some contrasts
Fixed effects and functional form	Additive unit effects and common time shocks; correct lag structure; a linear, homogeneous CET if one coefficient is to have that interpretation	Fixed effects need not remove the omitted process; the coefficient may be only a best linear approximation
Finite-T estimation	A time dimension or correction adequate for bias from within-transforming lagged outcomes and other predetermined regressors	The population CET may be identified while the conventional within estimator has Nickell-type finite-T bias
Inference	An uncertainty procedure appropriate for residual serial and cross-sectional dependence	The population target can remain unchanged while standard errors and confidence intervals are invalid

The distinction between population identification and estimation matters in dynamic panels. With lagged outcomes, demeaning generally correlates the transformed lag with the transformed innovation, producing Nickell-type bias at finite T even when the level conditional mean identifies β_{CET} (Nickell, 1981). Serial and cross-sectional dependence are separately inference problems when the remaining innovations are conditionally mean zero; an omitted common factor correlated with treatment is instead an identification problem.

For short panels, bias corrections or Arellano–Bond may be useful alternatives under their own moment and instrument conditions, but GMM is not uniformly superior and does not repair a failed causal clock (Arellano and Bond, 1991). These qualifications align the benchmark with the sequential-identification logic in the dynamic-panel literature (Blackwell and Glynn, 2018; Imai and Kim, 2021).

5 Benchmarking Dynamic Specifications for the CET

This section develops the paper's main applied implication of the IVB framework. For the CET, the default applied comparison should be between dynamic specifications that differ in whether they condition on contemporaneous Z_t, not between a dynamic model and a static TWFE short regression treated as a causal gold standard. The benchmark is therefore not a substitute for IVB. It is the most important operational payoff of IVB in the TSCS setting studied here.

5.1 Collider and Dual-Role Controls in TSCS

We begin with the dual-role setting in which Z_{t-1} is a confounder for (D_t, Y_t), but Z_t is contemporaneously affected by both treatment and outcome. In this structure, a variable such as GDP can serve as a lagged state variable while its contemporaneous value has a different causal role.

The dual role is visible in Figure 3. Without conditioning, $D_t \leftarrow Z_{t-1} \rightarrow Y_t$ is an open backdoor path. Conditioning on Z_{t-1} closes that fork but opens the inherited path $D_t \leftarrow D_{t-1} \rightarrow Z_{t-1} \leftarrow Y_{t-1} \rightarrow Y_t$, because Z_{t-1} is a collider on that path. Adding Y_{t-1} blocks this inherited path at the noncollider Y_{t-1} and also blocks the distinct omitted-dynamics path $D_t \leftarrow D_{t-1} \rightarrow Y_{t-1} \rightarrow Y_t$. This d-separation statement is conditional on the displayed graph: it does not close $D_t \rightarrow Z_t \leftarrow Y_t$ if Z_t is conditioned on, and it does not rule out other omitted paths or finite-T estimation bias.

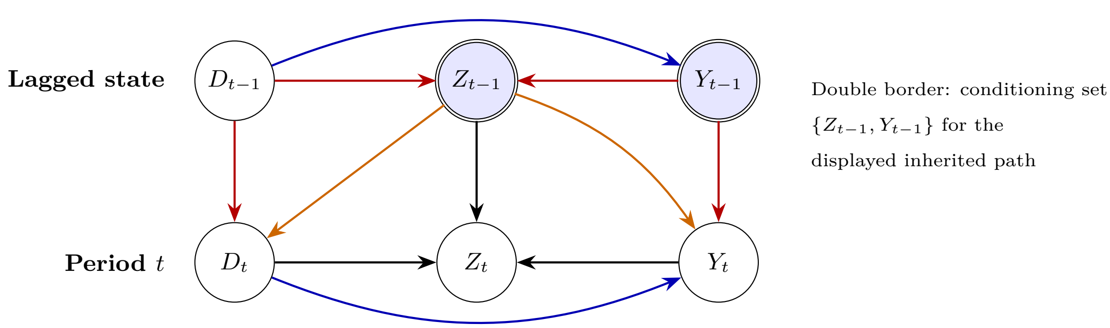

Figure 3: Dynamic dual-role DAG. Orange arrows form the lagged-confounding fork; red arrows trace the inherited collider path. Conditioning on Z_{t-1} blocks the former but opens the latter. Conditioning additionally on Y_{t-1} blocks that inherited path, subject to the displayed DAG and the paper's timing and identification conditions.

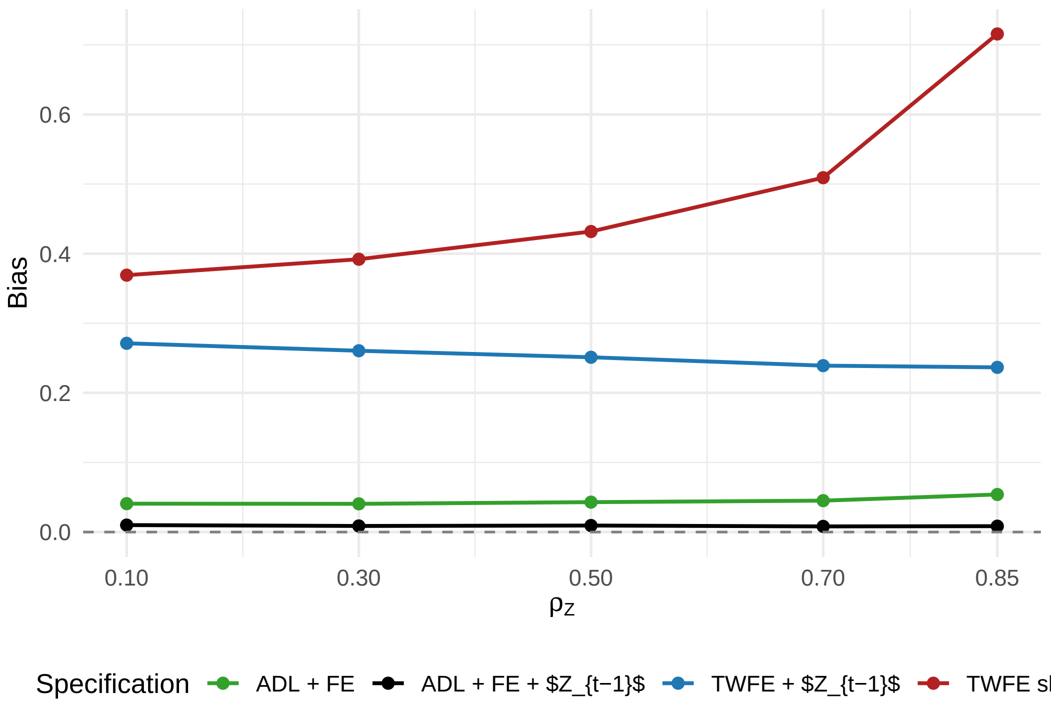

Figure 4: Dual-role simulation comparing static and dynamic specifications. In the main design, ADL + FE + Z_{t-1} keeps bias near zero across the reported range of collider persistence, while the static TWFE specifications retain appreciable bias.

In the main dual-role design, the bias of ADL + FE + Z_{t-1} stays between 0.008 and 0.010, while the bias of TWFE + Z_{t-1} remains between 0.237 and 0.271. The static short model has the largest reported bias, reaching 0.716 in the most persistent scenarios. For this DGP, the relevant comparison is therefore between specifications that represent Z as a lagged state inside the dynamic outcome model and specifications that omit part of that state.

In the displayed DGP, outcome persistence transmits both the inherited collider path and the ordinary lagged treatment–outcome path through Y_{t-1}. Table 4 shows that when this edge is removed by setting $\rho_Y = 0$, the TWFE bias in the reported design is approximately zero. Conditioning on Y_{t-1} blocks both displayed paths when $\rho_Y > 0$, subject to the graph and scope conditions stated above; the simulation contrast does not identify how much of the improvement comes from each path.

This finding disciplines the paper's recommendation. The lagged outcome is not a general fix. In the displayed DGP, it conditions on the persistent outcome state through which both cited lag paths reach Y_t. The practical benchmark for this dual-role design is therefore ADL + FE with Y_{t-1} and substantively justified lags of candidate controls, subject to the identification and estimation conditions above.

Table 4: Bias by outcome persistence in the dual-role DGP. When $\rho_Y = 0$, the reported bias from conditioning on Z_{t-1} is approximately zero even without Y_{t-1}. When $\rho_Y > 0$, including Y_{t-1} conditions on the displayed persistent outcome state. This comparison does not decompose the ordinary dynamic and inherited-collider paths.

Outcome persistence (ρ_Y)	TWFE + Z_{t-1}	ADL + FE + Z_{t-1}
0.0	0.001	0.008
0.5	0.251	0.009
0.8	0.261	0.004

5.2 Over-Control with Contemporaneous Mediators

The second recurring problem is not collider bias but over-control. When Z_t lies on the path from D_t to Y_t, conditioning on Z_t shifts the estimate toward the direct effect. For the CET, the right benchmark is again dynamic: compare an ADL specification that leaves the contemporaneous mediator out of the regression with one that conditions on Z_t, and ask whether a lagged version Z_{t-1} can enter the benchmark without blocking the contemporaneous path.

Table 5: Over-control simulations for the CET. In both the pure-mediator and mediator-plus-confounder designs, conditioning on contemporaneous Z_t pulls the estimate toward the direct effect, whereas conditioning on Z_{t-1} preserves the contemporaneous total effect.

Scenario	Target total	Target direct	ADL total	ADL bad (Z_t)	ADL safe (Z_{t-1})
Pure mediator, $\rho_Z = 0.3$	1.20	1	1.201	0.998	1.199
Pure mediator, $\rho_Z = 0.7$	1.20	1	1.199	0.992	1.197
Mediator + confounder, $\rho_Z = 0.5$	1.03	1	1.091	1.023	1.031
Mediator + confounder, $\rho_Z = 0.7$	1.03	1	1.111	1.016	1.028

The clean mediator design is the simplest illustration. ADL total recovers the true contemporaneous total effect of about 1.20, ADL bad (Z_t) lands near the direct effect of 1.00, and ADL safe (Z_{t-1}) remains essentially equal to the total effect. The general design strengthens the same conclusion. When Z_{t-1} is also a confounder, including it improves the estimate by removing lagged omitted variable bias without blocking the contemporaneous path $D_t \to Z_t \to Y_t$.

This is why TWFE short is not the right benchmark for the mediator case. A static specification without the relevant dynamics mixes over-control with omitted dynamics, making the comparison harder rather than easier to interpret. The operational contrast should instead be ADL total versus ADL bad versus ADL safe.

5.3 Practical Benchmark

The evidence above yields a simple rule for applied TSCS work centered on the CET:

1. Start from an ADL + FE specification with lagged state variables, including Y_{t-1} and any substantively

justified lagged controls.

2. Treat any contemporaneous time-varying covariate Z_t as a potential bad control until the DAG rules out collider and mediator status.
3. Use the IVB decomposition to quantify the shift in the treatment coefficient when a candidate control is included.
4. Interpret that shift through the DAG: collider, mediator, confounder, or dual-role variable.
5. Revisit the benchmark if the estimand is not the CET. A control that is pre-treatment for the CET need not be innocuous for long-run or cumulative effects.

Additional simulations sharpen the scope of this recommendation. When the outcome equation contains nonlinear effects of Z_{t-1} or level-dependent trends of the kind emphasized in the recent DID literature, the bias of ADL(all) never exceeds 0.4 percent of β across stable scenarios. These are therefore not practical boundary conditions for the benchmark in our TSCS setting. By contrast, ADL(all) does not resolve contemporaneous unobserved confounding. In simulations where an unobserved variable affects both D_t and Y_t in the same period, the bias of ADL(all) ranges from 8 to 53 percent of β, while an oracle specification that observes the confounder keeps bias below 0.3 percent. The benchmark therefore addresses included-variable bias and related timing problems, not classical endogeneity from unobserved contemporaneous confounding.

This benchmark is intentionally narrower than a universal recommendation for all panel estimators. It is a defensible default for the specific but common problem the paper studies: deciding what to do with observed time-varying controls when the goal is the contemporaneous treatment effect.

6 Applications

The applications serve two purposes. First, they show that IVB yields interpretable empirical diagnostics rather than only theoretical algebra. Second, they show how the TSCS benchmark disciplines the interpretation of those diagnostics. Across 14 collider candidates in six studies, the median |IVB/SE| is 0.13, and only 1 case exceeds one standard error. That pattern is consistent with the simulation evidence: many applied settings contain candidate bad controls, but dynamic benchmarking and careful timing often keep the practically relevant distortion modest.

6.1 Application 1: Democracy and Ethnic Inequality

Leipziger (2024) estimates the effect of democratization on ethnic inequality with a two-way fixed-effects design in which democracy and GDP per capita are both lagged one calendar year relative to the outcome.

Indexing by the outcome year, the estimated exposure contrast is therefore $D_{t-1} \to Y_t$; equivalently, if $s = t-1$ denotes the exposure year, it is $D_s \to Y_{s+1}$. This application is an audit of a lagged exposure contrast, not a direct demonstration of the manuscript's core contemporaneous $D_t \to Y_t$ CET. The comparison remains useful because the same timestamp questions determine whether GDP closes before the democracy exposure in year s and because the coefficient shift shows how sensitive the lagged contrast is to conditioning on GDP. GDP is pre-outcome, but not demonstrably pretreatment relative to democracy: the two predictors share the same annual index, and the replication documentation does not provide within-year transition and measurement dates. We therefore classify the timing of GDP as ambiguous for the treatment contrast, rather than treating its lag label as dispositive. The practical question is whether GDP should be treated as a confounder, a treatment-responsive variable, or a dual-role variable.

GDP per capita is a plausible dual-role variable in this setting. Democratization can raise income over time (Acemoglu et al., 2019; Papaioannou and Siourounis, 2008), while ethnic inequality can depress income through weaker public-goods provision and conflict channels (Grundler and Link, 2024). The IVB decomposition therefore does not mechanically tell us to drop GDP. It tells us what the treatment estimate pays for including it.

Table 6: IVB decomposition for Leipziger (2024), public-services inequality measure.

Component	Estimate
$\hat{\beta}_{\text{short}}$	-0.0407
$\hat{\beta}_{\text{long}}$	-0.0352
$\hat{\theta}^{\star}$	-0.0497
$\hat{\pi}$	0.1123
$\text{IVB} = -\hat{\theta}^{\star}\hat{\pi}$	0.0056
$\lvert\text{IVB/SE}\rvert$	0.48
$\lvert\text{IVB}\rvert/\lvert\hat{\beta}_{\text{long}}\rvert$	15.9%

The estimated IVB is 0.0056, or about 0.48 standard errors. The treatment effect is attenuated by roughly 16 percent, but the shift remains well within ordinary sampling uncertainty. This is a useful example of the paper's practical stance: a candidate dual-role control can matter substantively without overturning the empirical conclusion.

6.2 Application 2: Postal Infrastructure and Economic Growth

Rogowski et al. (2022) provide the complementary case. Their panel is organized in five-year steps, the outcome is led one panel period, and the postal-stock and GDP measures enter the same regression row. Letting s index the baseline five-year row, the estimated exposure contrast relates the cumulative postal stock

D_s to forward growth Y_{s+1}, where $s+1$ is the next five-year panel step. This application is therefore an audit of a forward exposure contrast, not a direct demonstration of the manuscript's core contemporaneous $D_t \to Y_t$ CET. It remains useful for timing and sensitivity because a cumulative stock makes prior exposure explicit and because the coefficient shift reveals how strongly the forward contrast depends on conditioning on baseline GDP. Baseline GDP is a plausible state variable for subsequent growth, but it is not demonstrably prior to the treatment represented by an accumulated stock of post offices; it may already embody responses to earlier postal investment. Its timing is ambiguous for the treatment contrast even though it precedes the measured outcome window. Here the IVB formula reveals a large arithmetic shift but cannot decide, by itself, whether dropping GDP improves identification.

Table 7: IVB decomposition for Rogowski et al. (2022).

Control as z	$\hat{\beta}_{\text{short}}$	$\hat{\beta}_{\text{long}}$	$\hat{\theta}^{\star}$	$\hat{\pi}$	IVB	|IVB/SE|	IVB/$\hat{\beta}$ (%)
Log GDP p.c.	0.0083	0.0198	-4.0130	0.0029	0.0115	2.11	58.0
Log population	0.0206	0.0198	-1.9465	-0.0004	-0.0008	0.15	-4.0
Urbanization	0.0192	0.0198	-3.1421	0.0002	0.0006	0.11	3.0
Polity2	0.0191	0.0198	0.0341	-0.0221	0.0008	0.14	3.8

For lagged GDP per capita, $|\text{IVB/SE}| = 2.11$, the largest value among the published applications. But this is precisely the case in which the formula must not be overread. GDP's causal role is unresolved: conditional convergence makes it a possible confounder; inherited postal infrastructure may make it a legacy treatment-responsive state or mediator; and both channels could give it a dual role. The large coefficient shift is therefore descriptive until a DAG and evidence on the relevant measurement and exposure windows establish more. The lesson is not "drop GDP.' ' The lesson is that a large specification shift demands explicit causal reasoning about timing and role. This is exactly the use case for IVB in a paper whose applied payoff is a benchmark rather than an automatic decision rule.

7 Conclusion

This paper introduces included variable bias as a general diagnostic problem that arises when researchers condition on observed covariates that may themselves respond to treatment. In the linear nested models used throughout TSCS work, IVB admits a simple closed-form decomposition. That decomposition does not decide whether a control is admissible; DAGs and timing do that work. But it does quantify the size and direction of the resulting specification shift in a way that existing discussions of time-varying controls typically do not.

The paper's main applied payoff is narrower and more concrete. For TSCS settings focused on the contemporaneous treatment effect, the IVB framework implies a practical benchmark: ADL + FE with lagged state variables. That benchmark is not a universal solution for all panel estimators, nor is it a claim that lagged controls are always innocuous. It is a defensible starting point for the specific setting studied here. The simulations show why it works in the cases most likely to worry applied researchers: dual-role controls, inherited collider paths, and contemporaneous mediators.

This combination of a general diagnostic and a specific benchmark is the paper's contribution. Without IVB, the paper would risk collapsing into a familiar recommendation to use dynamic specifications. Without the benchmark, IVB would risk remaining an algebraic curiosity. Put together, they provide both a language for the problem and an operational default for a common applied setting.

The limits are equally important. The benchmark is local to the CET. A lagged covariate that is pre-treatment for $D_t \to Y_t$ need not be harmless for distributed, cumulative, or long-run effects. Additional simulations further show that the benchmark remains robust to nonlinear effects of lagged controls and level-dependent trends, but not to persistent unobserved contemporaneous confounding. And although the IVB idea may extend beyond the current domain, this paper does not claim results for synthetic-control-style estimators, synthetic DID, or factor models. Extending IVB to those settings is an important agenda for future work. The present paper establishes the idea and its main payoff in the linear TSCS case.

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